

Code-Layout, Readability and Reusability

Date	20 July 2026
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability

Code Layout Checklist

S.No	Code Quality Parameter	Description	Followed (Yes/No/Partial)	Remarks
1	Consistent Indentation	Uniform 4-space PEP8 style throughout	Yes	
2	Proper File Structure	Training pipeline (<code>train_model.py</code>), web app (<code>app.py</code>), templates, and static assets kept separate	Yes	
3	Meaningful Variable Names	e.g. <code>feature_cols</code> , <code>X_train</code> , <code>best_model</code> , <code>prediction_text</code>	Yes	
4	Function Names	e.g. <code>load_data()</code> , <code>clean_data()</code> , <code>remove_outliers_iqr()</code> , <code>train_and_evaluate()</code>	Yes	
5	Code Comments / Docstrings	Module-level docstrings in both <code>app.py</code> and <code>train_model.py</code> explain purpose and pipeline steps	Yes	
6	Modular Design	Each pipeline stage (cleaning, outlier removal, plotting, clustering, evaluation) is its own function	Yes	
7	No Redundant Code	Evaluation logic shared across both candidate models via <code>evaluate_model()</code>	Yes	
8	Error Handling	Flask <code>/predict</code> route wraps parsing/prediction in try/except and returns a friendly message instead of crashing	Yes	

Reusable Components / Modules

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level
1	<code>clean_data()</code>	Python / pandas	Called once in <code>train_model.main();</code> reusable for any retraining run on updated data	High
2	<code>remove_outliers_iqr()</code>	Python / pandas	Generic IQR-based outlier remover, works on any numeric column list	High

3	<code>evaluate_model()</code>	Python / scikit-learn	Reused for both Logistic Regression and Random Forest via the <code>candidates</code> dict in <code>train_and_evaluate()</code>	High
4	<code>elbow_method()</code>	Python / scikit-learn	Standalone K-Means/Elbow utility, reusable for any scaled feature matrix	Medium
5	Pipeline (StandardScaler + classifier)	scikit-learn	Both candidate models use the same scaler+classifier pipeline pattern	High

Overall Code Quality Assessment

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Clear separation between training pipeline and serving app
Readability	4	Descriptive naming and docstrings throughout
Reusability	4	Evaluation and preprocessing helpers are generic, not hard-coded to one model
Documentation / Comments	4	Module docstrings explain the full pipeline end-to-end
Overall Score	4.5/5	